# LLMs in the Study of Human Language Processing

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University of Tübingen

# Outline

Why LLMs to study human language processing

What can we extract from LLMs?

Types of questions LLMs let us ask

LLMs as research tools

# Why LLMs to Study Human Language Processing

- Accessible computational systems
- Text-trained predictive learners
- Unprecedented linguistic competence

# Outline

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LLMs as research tools

# I. Prompting

- Choices
- Ratings
- Continuations

# II. Probabilistic Expectations

- Next-word probability
- Surprisal
- Entropy / Uncertainty

# III. Internal Representations

- Hidden states
- Contextual embeddings
- Attention patterns

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# Behavioural Alignment with Humans

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- Sound symbolism
- Semantic illusion

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**Goal**

Test whether a model reproduces **behavioral effects** observed in psycholinguistic experiments.

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# Behavioural Alignment with Humans

## Structural priming

**Do large language models resemble humans in language use?**

**Zhenguang G. Cai[1,2], Xufeng Duan[1], David A. Haslett[1], Shuqi Wang[1], Martin J. Pickering[3]**
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# Behavioural Alignment with Humans

## Structural Priming

Humans tend to reuse recently encountered syntactic structures.

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**Prime sentence**

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(Double-object structure)

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**Target prompt**

*The patient showed ...*

**Human tendency**

*The patient showed* ***the nurse his hand****.*

# Linking / Correlational Inference

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**Examples**

- Surprisal $\rightarrow$ Reading times/N400
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- Entropy $\rightarrow$ Eye movements

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**Interpretation**
LLM measures may operationalize latent processing variables such as *predictability*, *compatibility*, and *uncertainty*.

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# Linking / Correlational Inference

## Surprisal/Embedding similarity → N400

SPECIAL ISSUE:
Cognitive Computational Neuroscience of Language

### Strong Prediction: Language Model Surprisal Explains Multiple N400 Effects

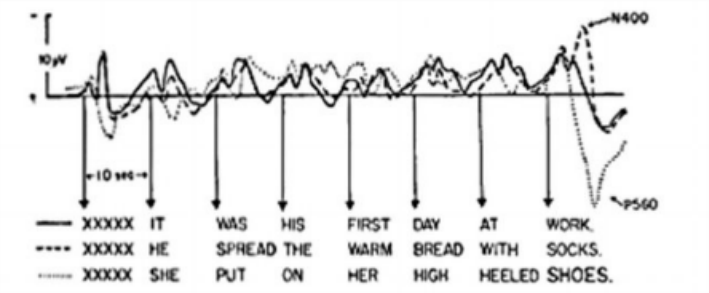
James A. Michaelov[1], Megan D. Bardolph[1], Cyma K. Van Petten[2], Benjamin K. Bergen[1], and Seana Coulson[1]

[1]Department of Cognitive Science, University of California, San Diego, La Jolla, CA, USA
[2]Department of Psychology, Binghamton University, State University of New York, Binghamton, NY, USA

**Keywords:** distributional semantics, ERPs, N400, neural language models, predictive coding

# Linking / Correlational Inference

## Surprisal/Embedding similarity → N400




Kutas & Hillyard (1980)

**Predictability Account**

Sentence Context

↓

Expectation about upcoming word

↓

Prediction Error

↓

N400 Amplitude

---

LLM Surprisal

(contextual probability)

**Similarity Account**

Sentence Context

↓

Activated Semantic Features

↓

Compatibility with incoming word

↓

N400 Amplitude

---

Embedding Cosine Similarity

(context-word compatibility)

# Sufficiency / Learnability Inference

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- Pragmatic implicature
- Agreement phenomena

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If a text-trained model captures it, strong modularity claims are **in-principle constrained**.

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## Syntactic island constraints

Using Computational Models to Test Syntactic Learnability

*Ethan Gotlieb Wilcox*
*Richard Futrell*
*Roger Levy*

# Sufficiency / Learnability Inference

## Syntactic island constraints

**Long-distance dependencies**

*I know* ***what*** *the lion devoured __ yesterday.*

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*I know what the lion devoured **the gazelle** yesterday.*

Unexpected object → increased processing difficulty.

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Unexpected object → increased processing difficulty.

**Island structures block the dependency**

*I know what the actress bought the painting*
*that depicted __ yesterday.*

# Model Comparison as Theory Testing

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**Examples**

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- RLHF vs. base models
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Architectural components that improve alignment become candidate **explanatory variables**.

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# Model Comparison as Theory Testing

## Multimodal vs. text-only

**Do Multimodal Large Language Models and Humans Ground Language Similarly?**

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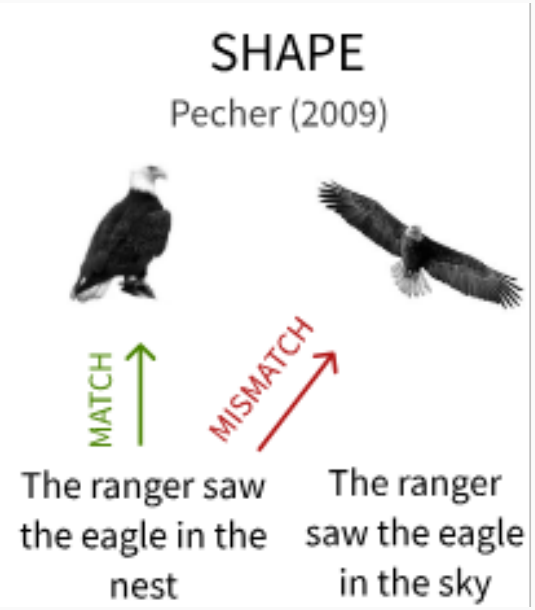
## Multimodal vs. text-only



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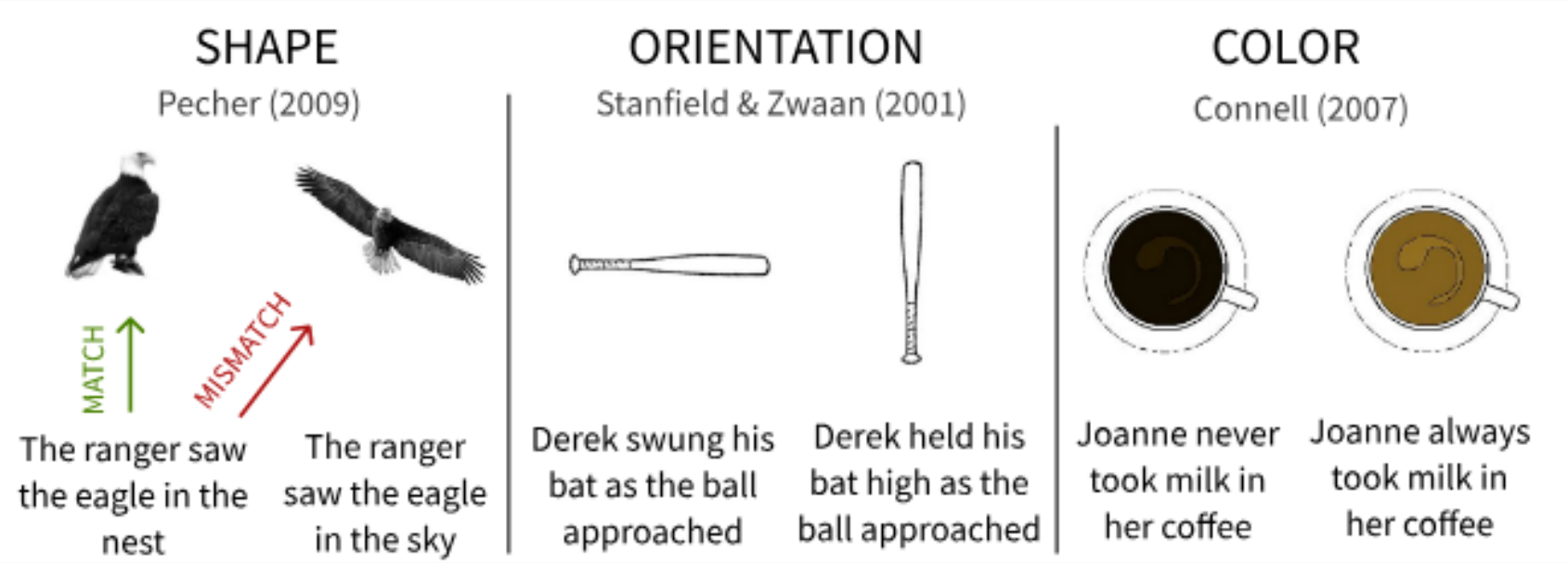

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Architectural components that improve alignment become candidate **explanatory variables**.

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## RLHF vs. base models

**Pragmatic Reasoning in GPT Models: Replication of a Subtle Negation Effect**

**Francesca Capuano (francesca.capuano@uni-tuebingen.de)**
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# Model Comparison as Theory Testing

## RLHF vs. base models

| | GPT-3 | InstructGPT | GPT-3.5 (Chat) | GPT-4 (Chat) | Humans |
|---|---|---|---|---|---|
| Experiment 3 (this) | neg >aff** | neg >aff** | n.s. | n.s. | n.s. |
| Experiment 6 (see) | n.s. | neg <aff* | neg >aff** | neg > aff* | neg >aff*** |
| Experiment 7 (want) | n.s. | n.s. | n.s. | neg > aff*** | neg >aff*** |

# Representation Questions

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- Probing classifiers
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These studies test whether abstract linguistic structure is **recoverable from internal representations**, informing debates about distributed vs. symbolic encoding.

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# Representation Questions

## Syntactic tree decodability

**A Structural Probe for Finding Syntax in Word Representations**

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**Christopher D. Manning**
Stanford University
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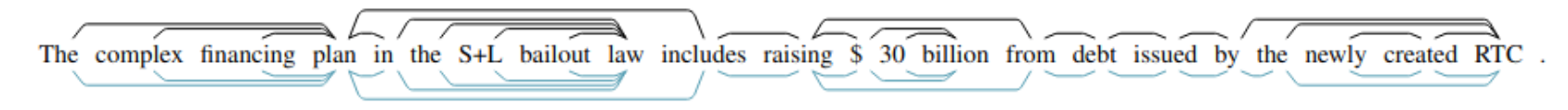
# Representation Questions

## Syntactic tree decodability



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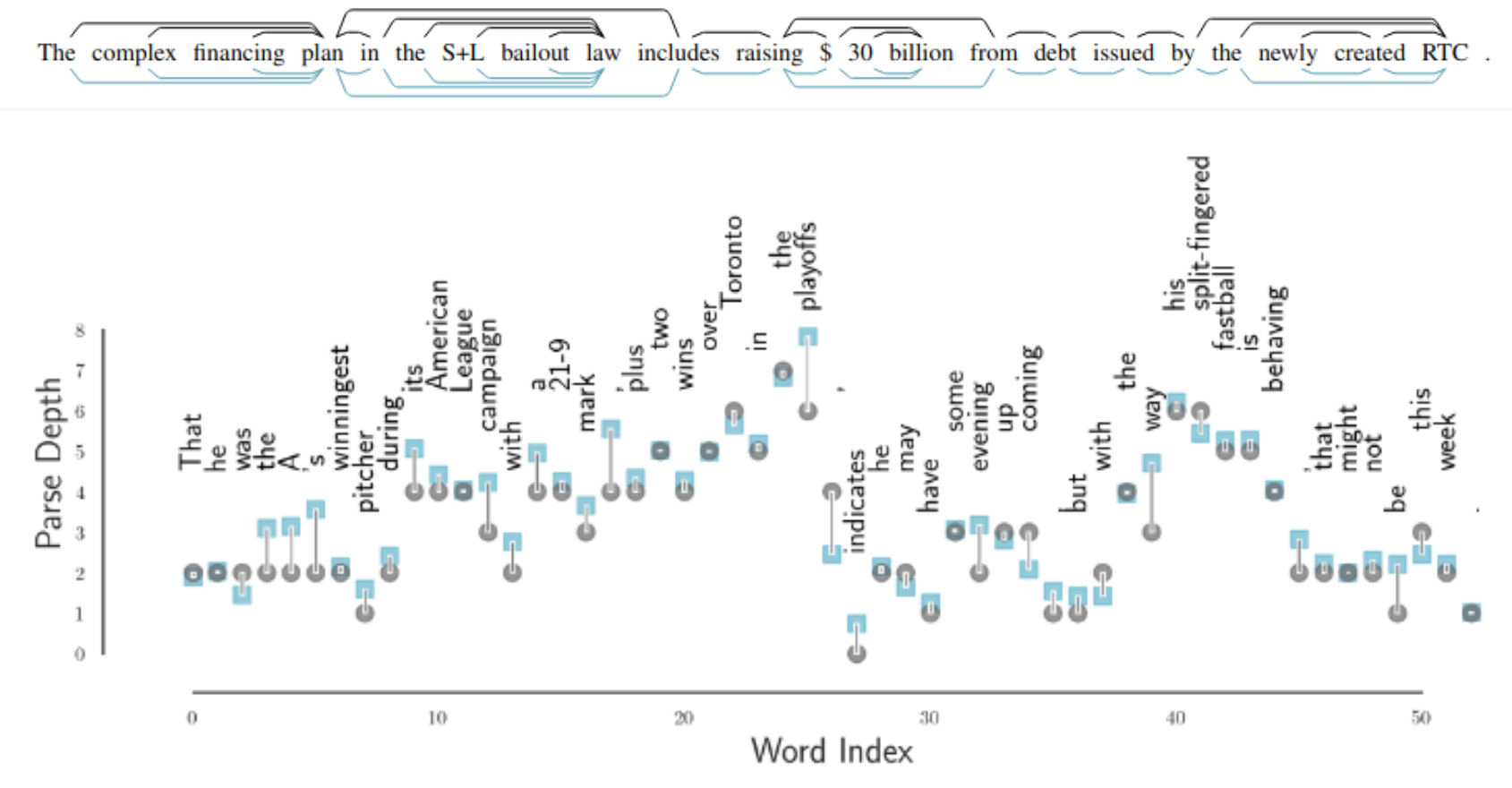

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**Examples**

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Mechanistic interpretability identifies the specific **computational circuits** that implement a behavior at the algorithmic level.

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- Induction heads
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Mechanistic interpretability identifies the specific **computational circuits** that implement a behavior at the algorithmic level.

## Attention circuits

# INTERPRETABILITY IN THE WILD: A CIRCUIT FOR INDIRECT OBJECT IDENTIFICATION IN GPT-2 SMALL

**Kevin Wang**[1], **Alexandre Variengien**[1], **Arthur Conmy**[1], **Buck Shlegeris**[1] **& Jacob Steinhardt**[1,2]
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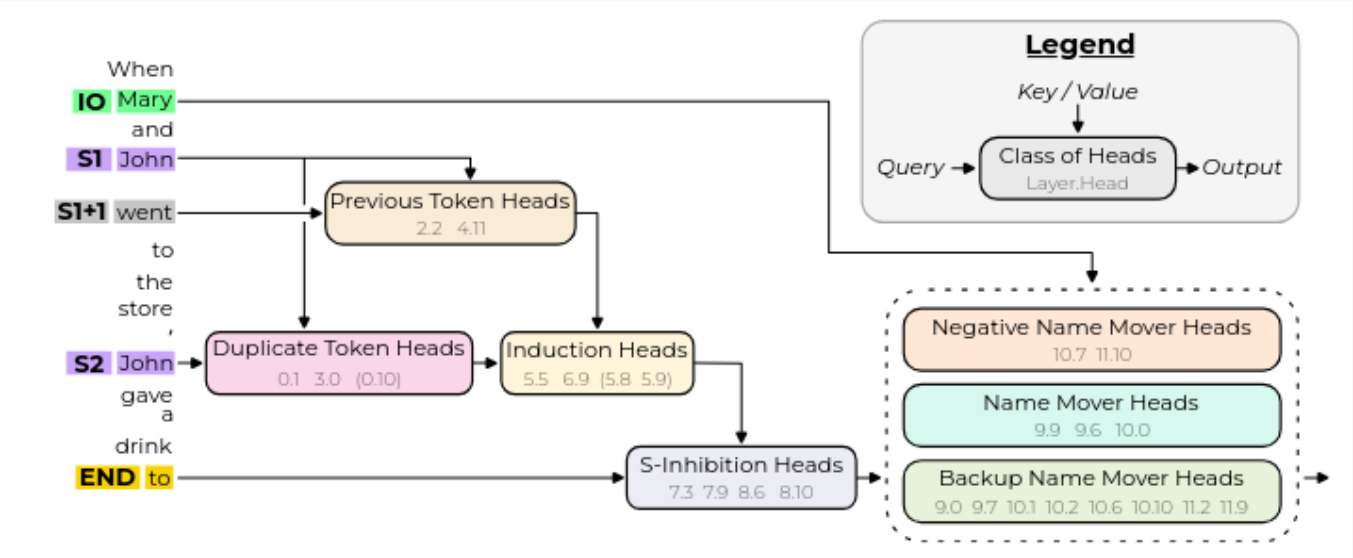
# Mechanistic Interpretability

## Attention circuits



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## Examples

- Norm augmentation
- Linguistic annotation
- Stimulus generation
- Experiment scripting

## Interpretation

LLMs can function as research tools without requiring direct cognitive interpretation.

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**Interpretation**
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# LLMs as Research Tools

## Norm augmentation

Behavior Research Methods (2024) 56:6082–6100
https://doi.org/10.3758/s13428-024-02337-z

ORIGINAL MANUSCRIPT

Check for updates

### Can large language models help augment English psycholinguistic datasets?

Sean Trott[1]

Accepted: 5 January 2024 / Published online: 23 January 2024
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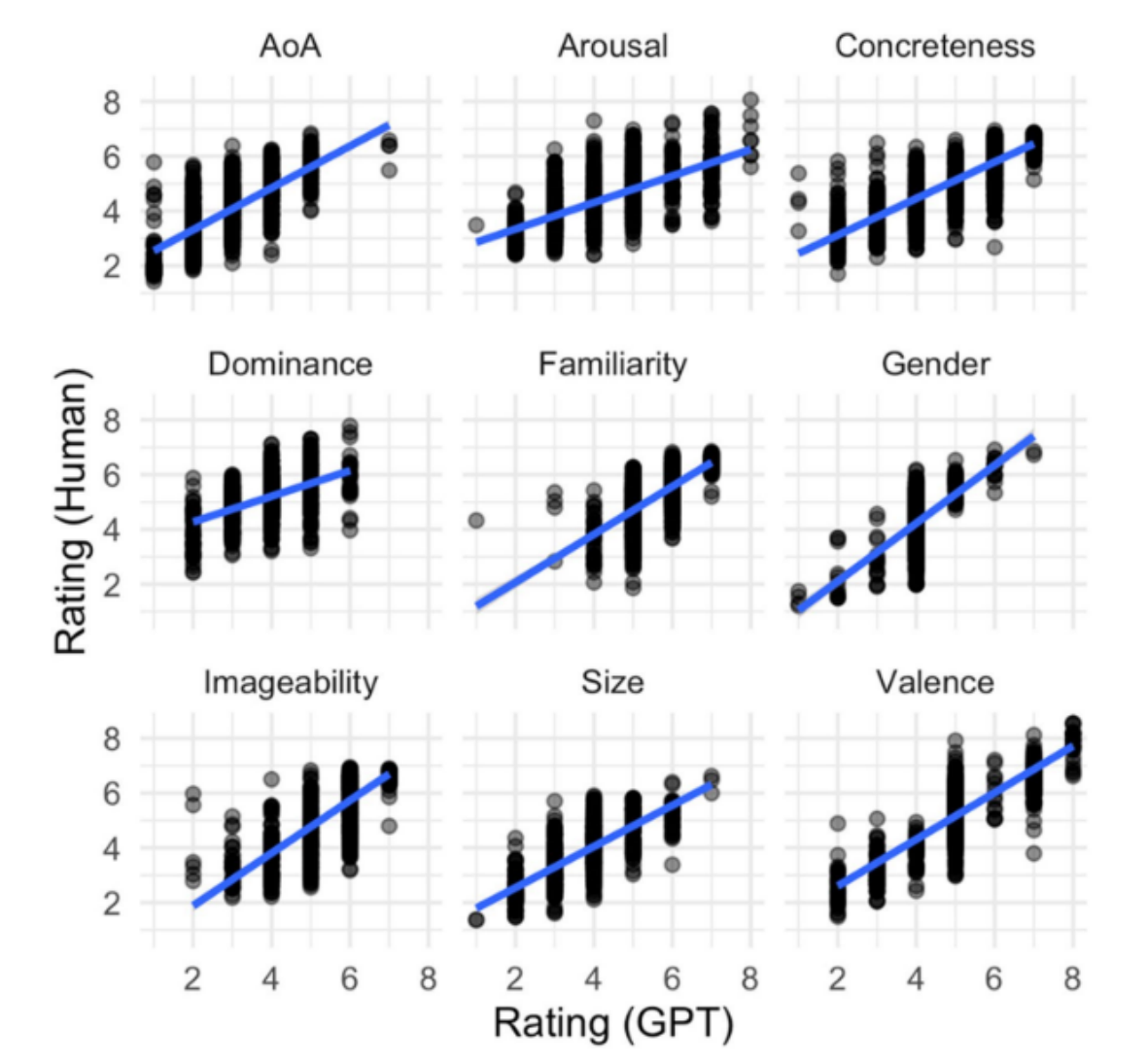
# LLMs as Research Tools

## Norm augmentation



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